Assignment 3.

1a

instructions	Issue	Disp	Ex	WB
fld f_1, 16(x12)	1	2	3-4	5
fld f_2, 32(x13)	2	3	5-6	7
fld f_3, 8(x14)	3	4	7-8	9
fmul.d f_4, f_1 f_2	5	8	9-12	13
fdiv.d f_5, f_4 f_8	7	18	19-38	39
fadd.d f_6, f_4, f_3	6	14	15-16	17
fsub.d f_7 f_6 f_4	8	14	17-18	19

instructions : 7

Total cycle : 39

$$CPI = \frac{\text{Total cycle}}{\text{instructions}} = \frac{39}{7} \approx 5.57$$

2a

instructions	Issue	Ex	WB
fld. f1 10(x12)	1	2-3	4
fld f2, 32(x13)	1	2-3	4
fld f3, 8(x14)	1	2-3	4
fmul.d f0 f1 f2	2	5-8	9
fdiv.d f5 f0 f8	5	13-32	33
fadd.d f8, f0, f3	3	10-11	12
fsub.d f7, f5 f0	4	10-11	12

2b

instructions 7

Total cycle 33

$$cpi = \frac{33}{7} \approx 4.71$$

2c)

$$speed = \frac{39}{33} \approx 1.18$$